GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Signals & Systems — Comprehensive Formula Sheet

Fourier Transforms, Laplace ROC, Z-Transform Pairs, LTI Convolution & Sampling Theorem

GATE Section: Section 2 • **Marks:** 7 - 9 Marks • **Core Focus:** Transform Pairs & LTI Properties

TRANSFORM PROPERTIES & SAMPLING SHORTCUTS
- **Convolution in Time = Multiplication in Frequency:** x(t) * h(t) <==> X(f) * H(f).
- **Multiplication in Time = Convolution in Frequency:** x(t) * y(t) <==> X(f) * Y(f).
- **Nyquist Rate:** For m(t) with max frequency fm, Nyquist rate = 2*fm. For m^2(t), Nyquist rate = 4*fm.

SECTION: ELEMENTARY SIGNALS & LTI SYSTEMS

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Continuous & Discrete Energy	**E = integral_{-inf}^{inf} |x(t)|^2 dt** **E = sum_{n=-inf}^{inf} |x[n]|^2**	Energy signal if 0 < E < inf, Power P = 0. Power signal if 0 < P < inf, Energy E = inf.	Periodic signals are always power signals! Average power = (1/T)*integral_T |x(t)|^2 dt.
Convolution Integral	**y(t) = x(t) * h(t) = integral_{-inf}^{inf} x(tau) h(t - tau) dtau**	Commutative: x*h = h*x Associative: x*(h1*h2) = (x*h1)*h2	Width property: Duration of y(t) = Duration(x) + Duration(h).
BIBO Stability	**integral_{-inf}^{inf} |h(t)| dt < inf (CT)** **sum_{n=-inf}^{inf} |h[n]| < inf (DT)**	Impulse response h(t) must be absolutely integrable / summable	In s-domain: All poles must lie in strictly Left-Half s-plane. In z-domain: All poles must lie strictly inside unit circle |z| < 1.
Causality Condition	**h(t) = 0 for t < 0 (CT)** **h[n] = 0 for n < 0 (DT)**	LTI system response cannot precede excitation	In s-domain: ROC is a right-half plane (sigma > sigma_max). In z-domain: ROC is exterior of circle (|z| > r_max).

SECTION: CONTINUOUS-TIME FOURIER TRANSFORM (CTFT)

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
CTFT & Inverse CTFT	**X(omega) = integral x(t) e^(-j*omega*t) dt** **x(t) = (1 / 2*pi) * integral X(omega) e^(j*omega*t) domega**	omega = 2*pi*f	Dirichlet conditions: x(t) absolutely integrable, finite extrema, finite discontinuities.
Duality Property	**If x(t) <==> X(omega), then X(t) <==> 2*pi*x(-omega)**	Ex: rect(t/T) <==> T*sinc(omega*T / 2) T*sinc(W*t / 2) <==> 2*pi*rect(omega / W)	Saves lengthy integration in sinc-to-rect and rect-to-sinc problems.
Time Scaling & Shifting	**x(a*t) <==> (1 / |a|) * X(omega / a)** **x(t - t0) <==> X(omega) * e^(-j*omega*t0)**	Time expansion = Frequency compression. Time delay adds linear phase delay -omega*t0.	Area under x(t) = X(0). Area under X(omega) = 2*pi*x(0).

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Differentiation in Time	**d^n x(t) / dt^n <==> (j*omega)^n * X(omega)**	Multiplies spectrum by (j*omega)^n	Used to compute transforms of piecewise linear waveforms by taking double derivatives.
Parseval's Energy Relation	**E = integral |x(t)|^2 dt = (1 / 2*pi) * integral |X(omega)|^2 domega**	Energy in time domain = Energy in frequency domain	Energy Spectral Density: ESD = |X(omega)|^2.

SECTION: LAPLACE & Z-TRANSFORM EQUIVALENCES

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Laplace Transform	**X(s) = integral_0^inf x(t) e^(-s*t) dt** **s = sigma + j*omega**	Unilateral Laplace transform	Initial Value: x(0+) = lim_{s->inf} s*X(s). Final Value: x(inf) = lim_{s->0} s*X(s) (valid only if s*X(s) poles are in LHP).
Standard Laplace Pairs	**delta(t) <==> 1** **u(t) <==> 1/s, ROC: Re(s) > 0** **e^(-a*t)*u(t) <==> 1/(s + a), Re(s) > -a** **cos(omega0*t) <==> s / (s^2 + omega0^2)**	t^n * e^(-a*t) <==> n! / (s + a)^(n+1)	Multiplication by t: t*x(t) <==> -dX(s)/ds.
Z-Transform	**X(z) = sum_{n=-inf}^{inf} x[n] * z^(-n)** **z = r * e^(j*Omega)**	Discrete-time frequency representation	Unit circle (r = 1) corresponds to DTFT. DTFT exists iff ROC includes unit circle |z| = 1.
Standard Z-Pairs	**delta[n] <==> 1, ROC: Entire z-plane** **u[n] <==> 1 / (1 - z^-1) = z / (z - 1), |z| > 1** **a^n * u[n] <==> z / (z - a), |z| > |a|** **-a^n * u[-n-1] <==> z / (z - a), |z| < |a|**	Same algebraic expression can have different ROCs!	Right-sided sequences have ROC: |z| > |a|; Left-sided sequences have ROC: |z| < |a|.
Sampling Theorem	**fs >= 2 * fm (Nyquist Rate)** **Ts <= 1 / (2 * fm) (Nyquist Interval)**	fm = maximum message frequency	If fs < 2*fm: Aliasing occurs. Reconstructed signal is corrupted. Anti-aliasing LPF required before sampler.